

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 17: Piston Engines Structure & 4-Stroke Cycle

Syllabus Source: CAE Oxford (pp. 1-48 (Pwr)) | Pass Target: >= 90% AviationExam

1. THE FOUR-STROKE OTTO CYCLE & VALVE TIMING

Aero piston engines operate on the 4-stroke constant-volume Otto cycle. One complete engine cycle requires four piston strokes (two complete 360° revolutions of the crankshaft / 720° total rotation):

| Stroke Name           | Piston Travel Direction                | Crankshaft Rotation & Valve Position                                                                                                                    |
|-----------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Induction          | TDC to BDC (Top to Bottom Dead Center) | 0° to 180°. Inlet valve open, exhaust valve closed. Low cylinder pressure draws in air-fuel mixture.                                                    |
| 2. Compression        | BDC to TDC                             | 180° to 360°. Both valves closed. Piston compresses mixture into combustion chamber (~7:1 to 9:1 compression ratio). Ignition fires ~20-25° BEFORE TDC! |
| 3. Power (Combustion) | TDC to BDC                             | 360° to 540°. Both valves closed. Burning fuel gases expand rapidly, driving piston down to generate mechanical crankshaft torque.                      |
| 4. Exhaust            | BDC to TDC                             | 540° to 720°. Exhaust valve open, inlet valve closed. Piston sweeps burned combustion gases out through exhaust manifold.                               |

- Valve Lead & Lag: Inlet valve opens before TDC (valve lead) and closes after BDC (valve lag) to capitalize on incoming gas inertia, maximizing cylinder volumetric filling.
- Valve Overlap: The period around TDC at the end of the exhaust stroke and start of the induction stroke when BOTH inlet and exhaust valves are open simultaneously. Incoming fresh charge assists scavenging of exhaust gases and cools valve heads.

2. POWER TERMINOLOGY, EFFICIENCIES & BMEP

Engine performance is measured across different efficiency boundaries:

- Indicated Horsepower (IHP): Theoretical power developed by expanding combustion gases inside the cylinders:  $IHP = (P_{LANK}) / 33,000$ , where P = Indicated Mean Effective Pressure (IMEP).
- Brake Horsepower (BHP): Actual usable mechanical power delivered at the propeller drive shaft, measured with a dynamometer:  $BHP = IHP - FHP$  (Friction Horsepower lost to bearing friction, oil pumping, and driving accessories).
- Mechanical Efficiency:  $BHP / IHP$  (typically 80% to 90%).
- Brake Mean Effective Pressure (BMEP): The average effective combustion pressure acting on the piston throughout the power stroke that produces the shaft BHP output.

3. SUPERCHARGING & TURBOCHARGING MECHANICS

Naturally aspirated engines lose power as altitude increases (~3% power loss per 1,000 ft) due to decreasing atmospheric density. Forced induction restores manifold pressure (MAP):

- Supercharger (Mechanically Driven): Centrifugal compressor driven by engine crankshaft gears. Draws significant engine power (parasitic loss) to spin compressor.
- Turbocharger (Exhaust Driven): Exhaust gas spins an exhaust turbine wheel connected via a common shaft to a centrifugal compressor. Harnesses normally wasted exhaust gas energy with minimal parasitic shaft load.
- Wastegate Controller: A butterfly or poppet valve that modulates how much exhaust gas passes across the turbo turbine vs bypassing directly overboard. Modulated by an oil-pressure actuator to maintain selected manifold pressure.
- Critical Altitude: The maximum altitude at which a turbocharged engine can maintain its maximum rated sea-level manifold pressure (or rated BHP) with the wastegate fully closed. Above critical altitude, manifold pressure and power decline steadily with altitude.

#### [!] EXAM TRAP (AVIATIONEXAM) - Critical Altitude Definition & Behavior

- Definition: The maximum altitude at which a turbocharged engine can produce its maximum rated continuous power / sea level manifold pressure. - Below Critical Altitude: The wastegate is PARTIALLY OPEN and closes progressively as altitude increases to maintain constant MAP. - At Critical Altitude: The wastegate reaches its FULLY CLOSED position! - Above Critical Altitude: The wastegate cannot close any further; MAP and engine power drop with further climb exactly like a naturally aspirated engine!

### STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

#### [+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Otto Cycle & Turbocharger Critical Altitude

1. Four Strokes (720 deg crankshaft rotation): Induction, Compression, Power, Exhaust. 2. Spark Advance: Spark fires 20-25 deg Before Top Dead Center (BTDC) so peak combustion pressure occurs slightly after TDC (approx 12 deg ATDC). 3. Critical Altitude: The altitude where the turbocharger wastegate reaches 100% FULLY CLOSED. Above critical altitude, manifold pressure (MAP) decreases with increasing altitude.

### 4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

#### Q1: What is 'valve overlap' in a four-stroke aero piston engine?

- [A]: The time when both spark plugs fire together
- [B]: The angular period of crankshaft rotation around TDC when both the inlet and exhaust valves are open simultaneously
- [C]: The time when both valves are closed on the power stroke
- [D]: The clearance between valve rocker and stem

#### [?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Valve overlap occurs at the end of the exhaust stroke and start of the induction stroke when the inlet valve opens before the exhaust valve has completely seated, enhancing scavenging.

#### Q2: In a turbocharged aircraft piston engine, what is the 'Critical Altitude'?

- [A]: The altitude where detonation always occurs
- [B]: The maximum altitude at which the engine can maintain its maximum rated power or sea-level manifold pressure with the wastegate fully closed
- [C]: The altitude where the turbocharger compressor stalls

- [D]: The service ceiling of the aircraft

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Critical altitude is reached when the turbo wastegate is 100% closed. Below this altitude, full rated power can be maintained; above it, manifold pressure declines as ambient density drops.

**Q3: How does Brake Horsepower (BHP) compare mathematically to Indicated Horsepower (IHP)?**

- [A]: BHP is always greater than IHP
- [B]:  $BHP = IHP - \text{Friction Horsepower (FHP)}$
- [C]: BHP is unrelated to IHP
- [D]:  $BHP = IHP \times 1.5$

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Indicated Horsepower is the total chemical energy converted in the cylinders. Brake Horsepower is the net usable shaft power delivered to the propeller after deducting internal friction horsepower losses.